Deeper earthquake catalogs help physics-based forecasts and hurt learned ones

Aidan Yu

Earthquake catalogs are dominated by small events. Lowering a detection threshold from M2.5 to M1.0 multiplies the event count by roughly twenty, and the standard assumption is that those extra events improve forecasts of the large earthquakes people care about. The assumption is rarely tested directly, because changing the threshold normally changes everything else in the experiment at the same time.

What was measured

Two Southern California regions, two model classes, one controlled comparison. The target earthquakes, the spatial grid, the forecast windows and the scoring are identical across every run. The only variable is how deep the input catalog goes.

The two models are a fitted ETAS, the standard statistical triggering model behind operational aftershock forecasting, and a flexible neural density model with no physics built into it.

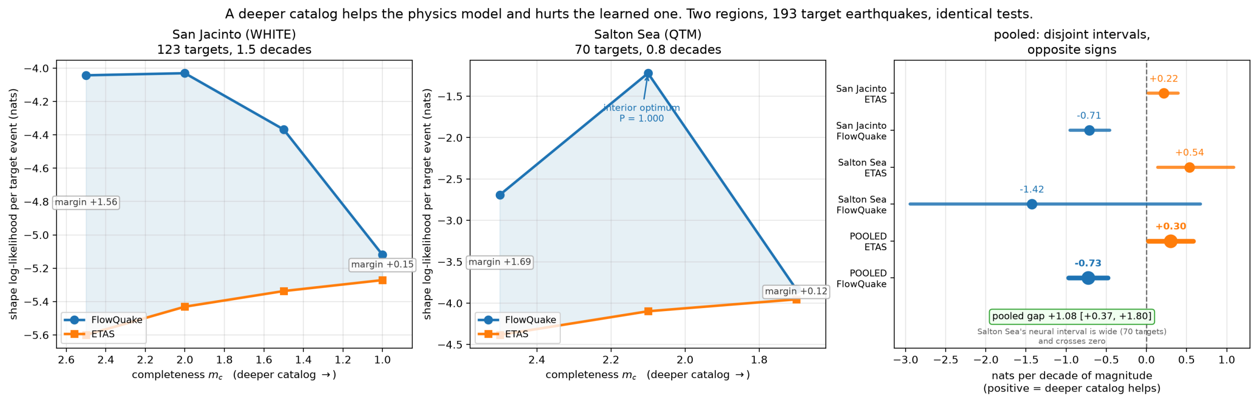
The result

The two run in opposite directions.

ETAS improves as the catalog deepens. It assigns about 1.35 times more probability to what happened, for every decade of magnitude added.

The neural model degrades. It assigns roughly half the probability, per decade.

Both are scored on the same target earthquakes, so the difference is a property of the models rather than of the data.



The same experiment in two regions. Blue is the neural model, orange is ETAS. Moving right means a deeper catalog, so more small earthquakes go in. ETAS climbs in both regions while the neural model falls, and the gap between them (shaded) nearly closes by the right-hand edge. The right panel shows the per-region and combined rates with 95% intervals. The two regions disagree about the middle of the curve: Salton Sea has a clear best threshold, San Jacinto does not.

The neural model is still the better forecaster at every threshold tested, in both regions. On a shallow catalog it leads ETAS by about five times. By the deepest catalog that lead has fallen to fifteen percent. It spends its whole advantage absorbing the extra data.

The obvious explanations do not hold

More than tripling the model size leaves the slope unchanged. The result is not an artifact of when training stopped, because every checkpoint across training was scored and the comparison uses the resulting plateau rather than a selected best. It replicates in a second region with different geology and a different set of target earthquakes.

A likely mechanism

This part is interpretation rather than measurement. The model trains on every event but is scored only on the rare large ones. Adding ten times more micro-earthquakes pulls the training objective toward the swarm, and the forecasts stop sharpening in space. ETAS cannot drift the same way, because its functional form is fixed in advance by physics, so extra events refine its conditioning without altering its shape.

What follows in practice

Detection threshold is a parameter with an optimum rather than a setting to push as far as the network allows. In one of the two regions that optimum is sharp. Moving from M2.5 to M2.1 improved the neural forecast substantially, and continuing to M1.7 cost more than twice what the first step gained. Anyone tuning a learned forecaster should search over the threshold instead of assuming deeper is better.

The effect should be largest for dense sensing systems. Distributed acoustic sensing on fiber optic cable detects well below the completeness of conventional networks, which is exactly the regime where this has not been measured. It is also the setting that could resolve the open question here.

The magnitude range available in these two regions spans one to one and a half decades, while a DAS catalog could span three or four, which would turn a slope into a full curve.

Compute

Training is not where the time goes. The models are small: about 15,000 parameters each, and 52,000 for the enlarged control. Training one of them for 12,000 steps takes three to four minutes on a single RTX 5090. All nine models across the three experiments come to roughly 35 minutes of training and 108,000 steps in total.

Evaluation is the expensive part. Scoring one model at one threshold means simulating forecasts across more than a thousand separate time windows at matched Monte Carlo resolution, and 297 of those runs sit behind the numbers above. The two most recent experiments, 165 of the runs, took about fourteen hours on the same 5090.

Compute is not the constraint here. Target earthquakes are.

Limitations

The result rests on 193 target earthquakes of M3 and above across two neighboring regions. That count, rather than compute or model choices, sets the precision, and only additional regions will improve it.

The two regions also disagree about the shape of the curve between its endpoints. One shows a clear optimum threshold and the other does not, and that difference is unexplained.

Rates are quoted per decade of magnitude added to the catalog, measured on 193 target earthquakes of M3 and above across two regions, with both models compared on identical target events.